

Sensor and Transducer

- ✧ **Sensor:** A device which provides a usable output in response to a specified measurand.

Examples:

- ✓ Temperature Sensor (e.g., Thermocouple)
- ✓ Pressure Sensor.
- ✓ Light Sensor.

- ✧ **Transducer:** A transducer is a device that converts *one form of energy into another*. Sensors are a type of transducer that specifically detect physical quantities.

Examples:

- ✓ Microphone: Converts sound waves into electrical signals.
- ✓ Loudspeaker: Converts electrical signals into sound waves.
- ✓ Piezoelectric Transducer: Converts mechanical pressure into electrical energy.

✧ **Difference between Sensor and Transducer**

Feature	Sensor	Transducer
Definition	A device which provides a usable output in response to a specified measurand.	A transducer is a device that converts <i>one form of energy into another</i> .
Function	Primarily used for <i>sensing and measuring physical parameters</i> (e.g., temperature, light, pressure).	Used for <i>energy conversion between different forms</i> (e.g., electrical to mechanical or vice versa).
Examples	Temperature sensor, pressure sensor, light sensor.	Microphone, loudspeaker, piezoelectric transducer.
Relation	All sensors are transducers.	Not all transducers are sensors (e.g., loudspeaker is a transducer but not a sensor).
Output	Usually an electrical signal (e.g., voltage, current).	Can be any form of energy (e.g., electrical, mechanical).

✧ **Sensor/transducers specifications:**

- 1) **Range:** The range of a sensor indicates the *limits between which the input can vary*. For example, a thermocouple for the measurement of temperature might have a range of 25-225 °C.
- 2) **Span:** The span is *difference between the maximum and minimum values of the input*. Thus, the above-mentioned thermocouple will have a span of 200 °C.
- 3) **Error:** Error is the *difference between the result of the measurement and the true value of the quantity being measured*. A sensor might give a displacement reading of 29.8 mm, when the actual displacement had been 30 mm, then the error is 0.2 mm.
- 4) **Accuracy:** Accuracy shows *how close a sensor's reading is to the true value*. For example, if a transducer has an accuracy of $\pm 1\%$ of its range (70 MPa), it means the reading could be off by ± 0.7 MPa.

- 5) **Sensitivity:** Sensitivity describes *how much a sensor's output changes in response to a change in input*. For instance, if a thermocouple's sensitivity is $41 \mu\text{V}/^\circ\text{C}$, its voltage will change by 41 microvolts for every 1°C change in temperature.
- 6) **Nonlinearity** Linearity is often specified in terms of percentage of nonlinearity, which is defined as: $\text{Nonlinearity (\%)} = \text{Maximum deviation in input} / \text{Maximum full scale input}$.

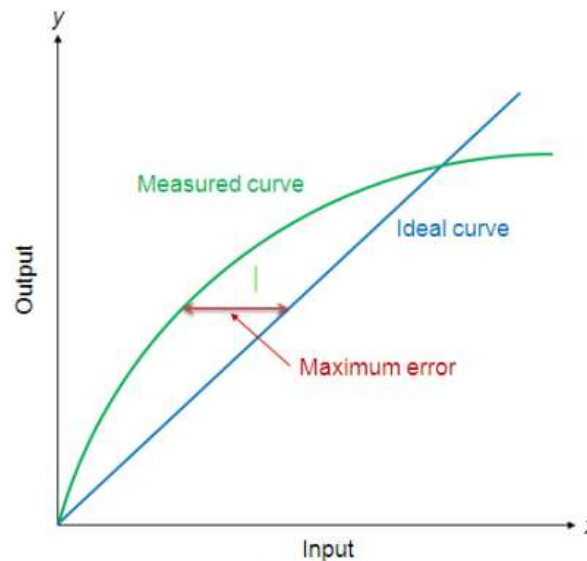


Figure 2.1.1 Non-linearity error

- 7) **Hysteresis:** Hysteresis is a type of sensor error. It occurs when the sensor *gives different output readings for the same input value*, depending on whether the input is increasing or decreasing. For example, when measuring temperature with a thermocouple, hysteresis can cause slight differences in readings during heating and cooling.

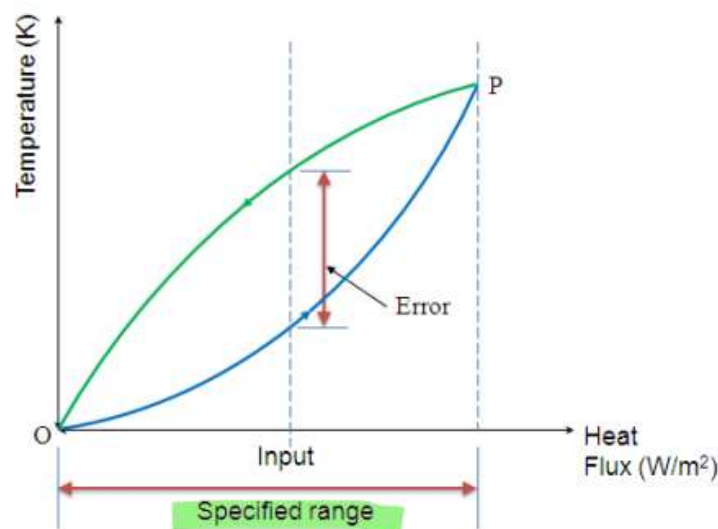


Figure 2.1.2 Hysteresis error curve

- 8) **Resolution:** Resolution is the smallest change in input that a sensor can detect.
Example: If a sensor measures up to 20 mm and shows changes as small as 0.2 mm, then its resolution is 0.2 mm.

9) Stability: Stability refers to a sensor's ability to provide the same output for a constant input over time.

Example: A stable temperature sensor will show the same reading if the temperature doesn't change.

10) Dead Band/Dead Time: Dead Band is the range where a sensor shows no response to small input changes. Dead Time is the delay between applying an input and seeing an output change.

Example: If a sensor doesn't respond to slight pressure changes between 0 and 2 psi, that's the dead band.

11) Repeatability: Repeatability is the ability of a sensor to give the same output when the same input is applied repeatedly.

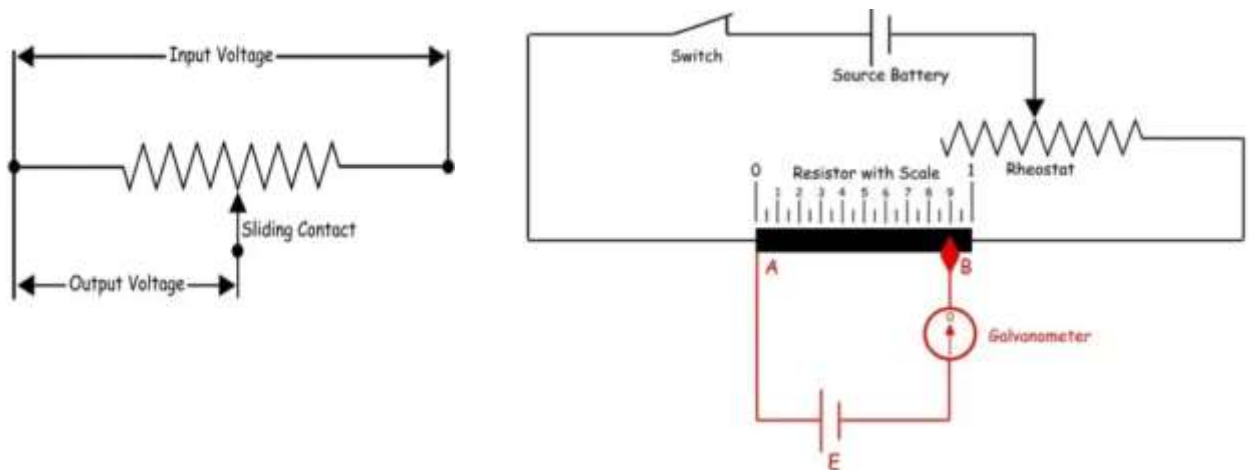
Example: A pressure sensor that gives 100 psi each time 100 psi is applied has good repeatability.

12) Response Time: Response Time is how quickly a sensor reacts to a change in the input.

Example: A thermometer that shows a temperature change within 2 seconds has a response time of 2 seconds.

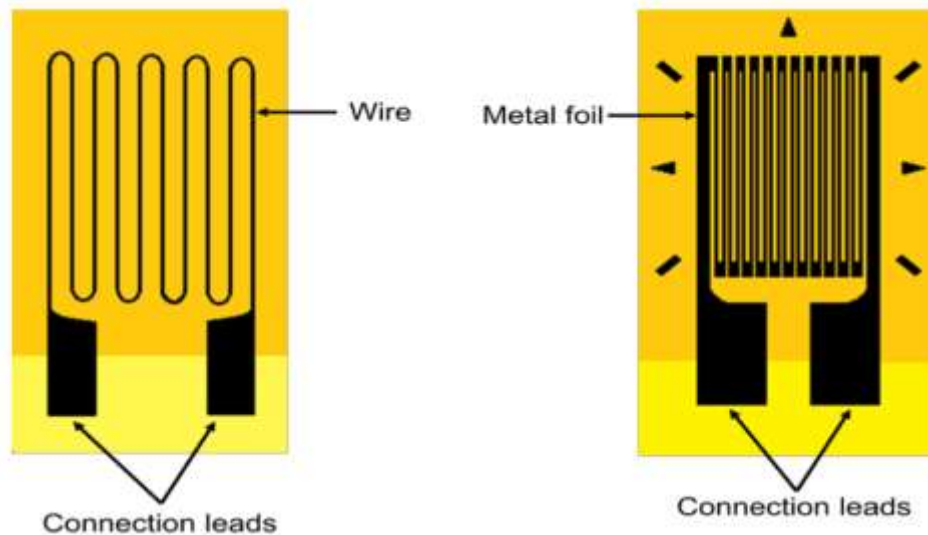
★ Potentiometer Sensors:

- A potentiometer is a *passive electronic component*.
- work by varying the position of a sliding contact across a uniform resistance
- the entire input voltage is applied across the whole length of the resistor, and the *output voltage is the voltage drop between the fixed and sliding contact*.
- Has the two terminals of the input source fixed to the end of the resistor. To adjust the output voltage the sliding contact gets moved along the resistor on the output side.
- Suppose we have connected two batteries in parallel through a galvanometer. The *negative battery terminals are connected together and positive battery terminals are also connected together through a galvanometer*.
- if the electric potential of both battery cells is exactly the same, there is no circulating current in the circuit and hence the galvanometer shows null deflection.



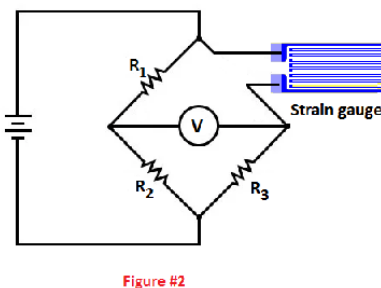
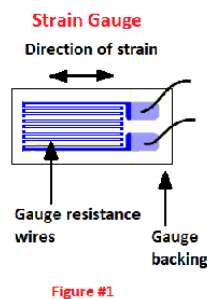
☼ Strain gauges:

- A strain gauge depends on the **electrical resistivity of any conductor**.
- The resistance in any conducting device is dependent on its length as well as the cross-section area.
- Suppose L_1 is the original length of the wire and L_2 is the new length after an external force is applied to it, the strain (ϵ) is given by the formula:
- $\epsilon = (L_2 - L_1) / L_1$

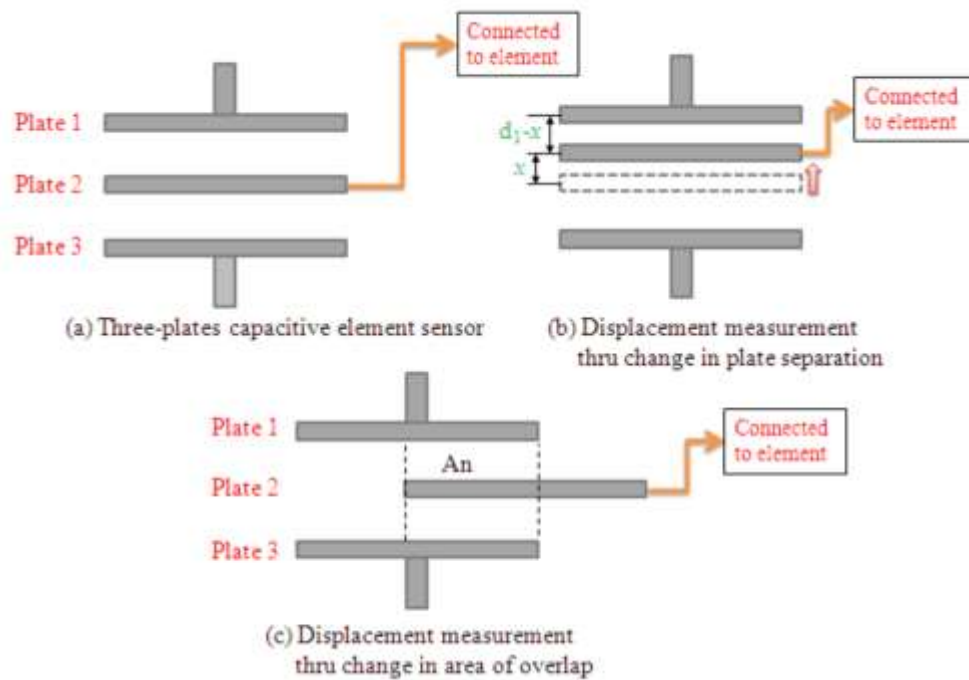


Whenever $R_1 / R_2 = R_4 / R_3$, the output voltage V_o is zero and the bridge is said to be balanced. Any change in the values of R_1, R_2, R_3 .

The output of the Wheatstone circuit is given by: $V_o = [(R_3 / (R_3 + R_4)) - (R_2 / (R_1 + R_2))] * V_{ex}$



✪ Capacitive element based sensor:



Points to Keep in Mind while Designing a Capacity Sensor

1. Maximize **capacitance with large-area, close-spaced plates**. Design electrode plates to measure the desired variable.
2. To handle lost capacitance from other circuits, surround the sensor with suitable guard or shield electrodes.
3. Calculate stray capacitance, output signal swing and sensor capacitance.
4. For *low noise*, choose a high enough excitation frequency. External and circuit-generated noise decreases when the excitation frequency increases.
5. Design circuit to provide immunity to environmental challenges and meet the accurateness of the specifications.

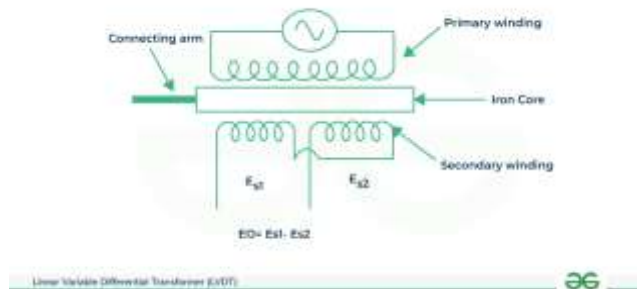
❖ **What is an LVDT? How does LVDT work step by step? Explain with LVDT construction diagram.**

The LVDT full form is “Linear Variable Differential Transformer” is LVDT. Generally, LVDT is a normal type of transducer. The main function of this is to *convert the rectangular movement of an object to the equivalent electrical signal*. LVDT is used to *calculate displacement and works on the transformer principle*.

The working of LVDT circuit diagram can be divided into three cases based on the position of the iron core in the insulated former.

Case 1: Core at Null Position

- ✓ Position: The core is centered between the two secondary windings (S1 and S2).
- ✓ Induced Voltages: The induced voltages in both windings (E_1 and E_2) are equal.
- ✓ Output Voltage: The output (E_{out}) is zero: $E_{out}=E_1-E_2=0$
- ✓ Meaning: No displacement has occurred.

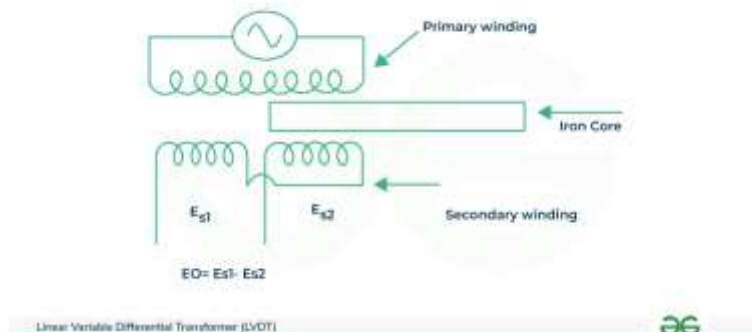


Case 2: Core Shifted to S1 from Null Position

- ✓ Position: The core moves to S1 from the center.
- ✓ Induced Voltages: The flux in S1 increases, while the flux in S2 remains less.
- ✓ Output Voltage: The output (E_{out}) is positive: $E_{out}=E_1-E_2>0$
- ✓ Meaning: Indicates a positive displacement to the S1.

Case 3: Core Shifted to S2 from Null Position

- ✓ Position: The core moves to S2 from the center.
- ✓ Induced Voltages: The flux in S2 increases, while the flux in S1 remains less.
- ✓ Output Voltage: The output (E_{out}) is negative: $E_{out}=E_1-E_2<0$
- ✓ Meaning: Indicates a negative displacement to S2.

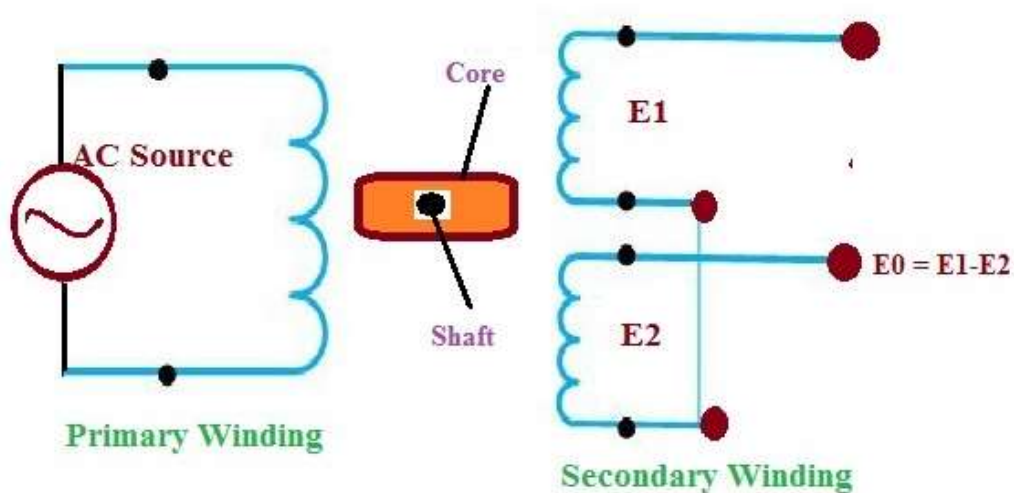


❖ **What is a RVDT? How does RVDT work step by step? Explain with RVDT construction diagram.**

The RVDT stands for **Rotary Variable Differential Transformer**. It is one kind of electromechanical transducer used to give the linear o/p which is proportional to the i/p angular displacement.

RVDT Construction

- **Components:** RVDT has a *primary winding* and *two secondary windings* on either side.
- **Core:** A soft iron core is attached to a shaft and can rotate between these windings.



Working Conditions:

Null Position:

- ✓ Position: The core is centered.
- ✓ Induced Voltages: The voltages in the secondary windings (E1 and E2) are equal but opposite.
- ✓ Output Voltage: The output (E0) is zero: $E0 = E1 - E2 = 0$

Clockwise Rotation:

- ✓ Position: The core rotates clockwise.
- ✓ Induced Voltages: E1 becomes greater than E2.
- ✓ Output Voltage: The output is positive: $E0 = E1 - E2 > 0$

Anticlockwise Rotation:

- ✓ Position: The core rotates anticlockwise.
- ✓ Induced Voltages: E2 becomes greater than E1.
- ✓ Output Voltage: The output is negative: $E0 = E1 - E2 < 0$